Training Intensity and Ambiguity Attitude Elicitation: Evidence from a Matching-Probability Task*

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Abstract: This study examines whether the amount of pre-task binary-choice training affects ambiguity attitudes elicited with the matching-probability method. In an online between-subjects experiment with a U.S. representative sample, participants complete either 5 or 100 practice binary choices before an otherwise identical elicitation task. I find no evidence that more extensive training significantly changes measured ambiguity attitudes. It does, however, reduce stochastic-dominance violations when the risky option dominates the ambiguous one, while leaving other quality indicators unchanged. These results suggest that, in this setting, the standard short training module is sufficient for ambiguity-attitude elicitation, and that heavier training yields only a narrow compliance gain at the cost of greater respondent burden.

Keywords: ambiguity; matching probability; training; binary choice; data quality

JEL Classification: D81; C91; D83

*The survey questionnaire was approved prior to implementation by the Ethics Commission of the Department of Economics at Gent University.

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1 Introduction

The matching-probability method is one of the leading tools for eliciting ambiguity attitudes in behavioral and experimental economics (Kahn and Sarin, 1988; Dimmock et al., 2016; Baillon et al., 2018b). In its common choice-based implementation, participants make a series of binary choices between an ambiguous lottery and a risky lottery, and the point of indifference identifies the matching probability of the ambiguous event. Because the task can be cognitively unfamiliar, experiments typically precede it with a short training module that introduces the stimuli, interface, and choice format (e.g., Abdellaoui et al., 2011; Attema et al., 2016; Abdellaoui et al., 2022). Yet it remains unclear whether the amount of such training affects what is ultimately elicited.

There are reasons to expect that it might. Training has been shown to influence the endowment effect (Plott and Zeiler, 2005) and elicited values in experimental auctions (Drichoutis et al., 2011; Corrigan et al., 2014), and repeated exposure to related decision problems often reduces choice anomalies in risky choice (e.g., Loomes et al., 2003; van de Kuilen and Wakker, 2006; DeKay et al., 2016). Under the discovered-preference view (Plott, 1996; Plott and Zeiler, 2005), additional practice may help respondents articulate preferences that are initially imperfectly formed. Applied to ambiguity elicitation, this suggests that more extensive training could either shift measured ambiguity attitudes or improve response quality.

This paper provides a direct test. In a between-subjects online experiment, participants complete either five or one hundred practice binary choices before an otherwise identical matching-probability task. The sparse condition represents standard practice; the extensive condition exposes participants to the full space of binary problems underlying the later elicitation task, providing a stringent test of whether additional training alters measured ambiguity attitudes.

I find no evidence that more extensive training changes measured ambiguity attitudes. Across elicited ambiguity-attitude indexes, estimated differences between conditions are small and statistically insignificant. At the same time, extensive training improves one narrow dimension of response quality: it reduces stochastic-dominance violations when the risky option transparently dominates the ambiguous one. By contrast, I find no comparable improvement in test-retest consistency, between-list monotonicity, or response time.

These results carry a practical implication for experimental design. In this task and sample,

the standard short training module appears sufficient for ambiguity-attitude elicitation via the matching-probability method. Heavier training yields only a limited gain in dominance compliance, while substantially increasing respondent burden.

2 Experimental Design

The experiment was conducted online on Prolific with a U.S. representative sample of 168 participants, of whom 82 were assigned to the *sparse* condition and 86 to the *extensive* condition. Each participant received a fixed participation fee of \$6. In addition, one participant in every ten was randomly selected to have one of her choices played out for real payment.[1]

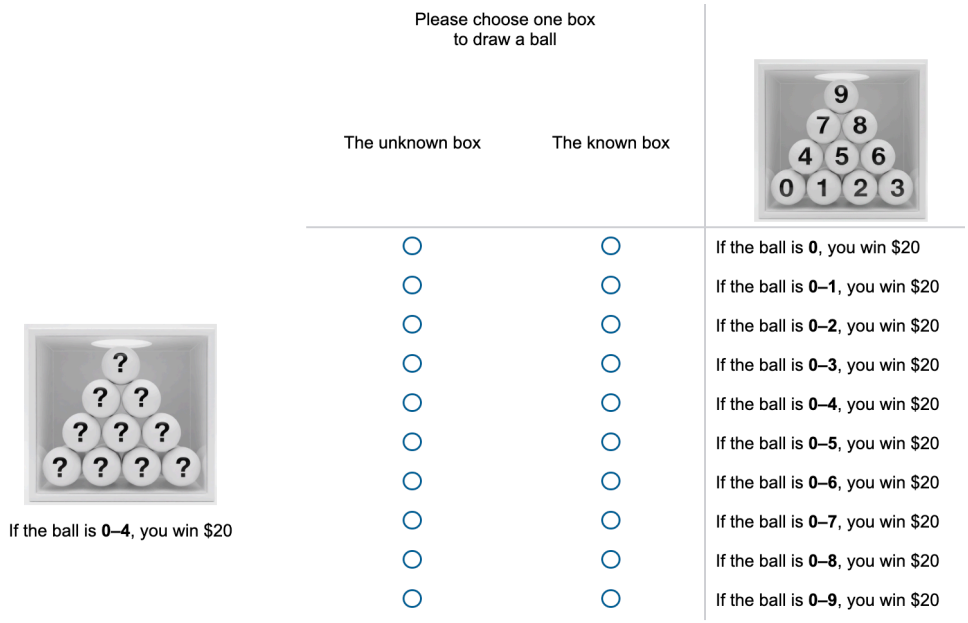
The main task elicits matching probabilities using a standard choice-list method (e.g., Baillon and Bleichrodt, 2015; Baillon et al., 2018a,b; von Gaudecker et al., 2025). In each list, participants choose between an ambiguous lottery on a fixed event E and a risky lottery whose winning probability varies across rows. The point at which a participant switches from the ambiguous lottery to the risky lottery identifies the matching probability $mp(E)$.[2] Figure 1a shows an example of the choice-list interface.

The elicitation uses a two-box design. Each box contains ten balls labeled 0 through 9. In the *risky* box, each number appears exactly once, so each draw is equally likely. In the *ambiguous* box, the composition is fully concealed, and participants are informed only that the distribution of numbers across the ten balls need not be uniform. For each box, ten winning ranges are defined — {0}, {0–1}, {0–2}, . . ., {0–9} — which together generate 100 binary decision problems. Participants complete 10 matching-probability choice lists, one for each winning range of the ambiguous box. Within a list, the ambiguous lottery is held fixed while the risky lottery's winning range expands across 10 rows, generating objective winning probabilities $p \in \{0.1, 0.2, \ldots, 1.0\}$. Both lotteries pay \$20 if the draw falls in the winning range and \$0 otherwise. List order is randomized across participants, and the direction of the within-list sequence (ascending or descending) is randomized between participants.

Before the matching-probability task, participants complete a training task consisting of

[1] To keep incentives comparable ex ante across conditions, the recruitment information in both conditions stated that the experiment would take approximately 30 minutes. The median duration was 20.0 minutes in the sparse condition and 30.3 minutes in the extensive condition.

[2] Single switching is not imposed. For lists with multiple switches, $mp(E)$ is approximated by the proportion of rows in which the ambiguous lottery is chosen (e.g., Holt and Laury, 2002).



(a) A matching-probability choice list example.

(b) A practice problem example.

Figure 1: Task interfaces.

standalone binary choices. Each training problem corresponds to a single row from the underlying problem space. This format is used for two reasons. First, it presents the same basic comparison between an ambiguous and a risky lottery that participants later face in the choice lists. Second, it avoids the memory and demand effects that could arise if full choice lists were used for training and later reappeared in the main task. Figure 1b shows an example of a training problem.

The only between-subjects manipulation is the amount of pre-task training. Participants in the *sparse* condition complete five randomly selected training problems, reflecting standard practice. Participants in the *extensive* condition complete all 100 problems, covering the full problem space underlying the later elicitation task. The training task always precedes the

matching-probability task, which is otherwise identical across conditions.

3 Results

3.1 Ambiguity Attitudes

Following conventional practice (Kahn and Sarin, 1988; Dimmock et al., 2016), I construct *event-specific* ambiguity attitude (AA) indexes. For each ambiguous event, the index $AA_{\hat{p}}$ captures the degree of "local" ambiguity aversion at the objective winning probability $\hat{p}$ that an ambiguity-neutral decision maker would assign to the event — hereafter the *ambiguity-neutral likelihood* (a-neutral likelihood for short). Formally,

$$AA_{\hat{p}} = \hat{p} - mp(\hat{p}),$$

where $mp(\hat{p})$ is the elicited matching probability for the ambiguous event with ambiguity-neutral likelihood $\hat{p}$. Positive values indicate ambiguity aversion, while negative values indicate ambiguity seeking.

Figure 2 plots the AA indexes by condition. In both conditions, participants are ambiguity-seeking for low likelihoods and ambiguity-averse for moderate to high likelihoods, consistent with the pattern reported in earlier studies. Most importantly, I find no evidence that more extensive training changes measured ambiguity attitudes. For all ambiguity-neutral likelihood levels, differences between the sparse and extensive conditions are statistically insignificant. The largest visible difference arises at likelihood 0.3, where the average AA index is −0.06 in the sparse condition and −0.10 in the extensive condition ($p = 0.26$).[3]

3.2 Response Quality

I next examine whether more extensive training improves response quality. The 100 binary-choice problems include 18 dominance checks: 9 in which the degenerate ambiguous lottery transparently dominates the risky lottery, and 9 in which the degenerate risky lottery transparently dominates the ambiguous lottery. Violations in these cases are unambiguous decision errors rather than preference expression. When the ambiguous lottery dominates, violation rates

[3] All comparisons between conditions use two-sided Mann–Whitney U tests. The results are unchanged when using two-sided t-tests.

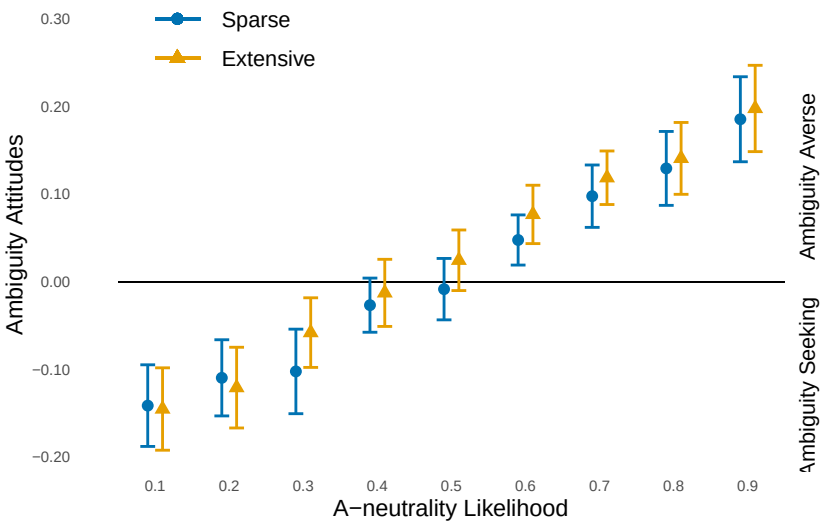


Figure 2: Event-Specific Ambiguity Attitude Indexes by Condition. *Notes*: Error bars denote 95% confidence intervals.

are similar across conditions (2.95 in the sparse condition versus 2.47 in the extensive condition; $p = 0.29$). By contrast, when the risky lottery dominates, participants in the extensive condition exhibit near-perfect compliance, averaging only 0.44 violations out of 9, compared with 2.16 in the sparse condition ($p < 0.01$). Extensive training therefore appears to improve one narrow aspect of response quality: identifying transparent stochastic dominance when the risky option dominates.

I also examine two further indicators of response quality. First, test–retest consistency is unaffected by the manipulation. For the repeated choice list administered at the end of the experiment, participants make 8.57 consistent choices on average in the sparse condition and 8.35 in the extensive condition ($p = 0.87$). Second, between-list monotonicity is identical across conditions, with an average violation frequency of 0.12 in both groups ($p = 0.75$). Finally, participants spend similar amounts of time on each matching-probability list: 18.1 seconds in the sparse condition and 21.5 seconds in the extensive condition ($p = 0.70$). Taken together, these results suggest that the gain from extensive training is limited to stochastic-dominance compliance in the risk-dominates direction, rather than reflecting a broader improvement in response quality.

4 Discussion

The finding that ambiguity-attitude measurement is robust to training intensity sits in some tension with the predictions of the related literatures reviewed above. One interpretation of the null effect is that preferences over ambiguous and risky lotteries are sufficiently clear that even minimal binary-choice exposure — five questions — is enough to activate them. This is supported by the fact that the majority of participants in both conditions report that the choice-list task is easy or very easy to understand. By this account, the marginal informational value of additional training diminishes rapidly: once respondents grasp the basic structure of the choice problem, further repetition does not meaningfully refine how they evaluate the ambiguous option. The pattern of stochastic dominance adherence is consistent with this view: the violation rate in the ambiguity-dominates direction — where correctly evaluating the ambiguous option is precisely what is at stake — remains unaffected by training intensity. This stands in contrast to the Becker-DeGroot-Marschak mechanism or experimental auctions, where the higher task complexity means that training depth shape elicited valuations (Plott and Zeiler, 2005; Drichoutis et al., 2011; Corrigan et al., 2014); the matching task imposes far lower cognitive and strategic demands, leaving less scope for training to alter measured attitudes.

Two considerations suggest the robustness of ambiguity-attitude measurement is conservative rather than fragile. University student samples are typically better educated and more mathematically sophisticated than the general population (Gillen et al., 2019), so if five training questions suffice in a representative online setting — which is also less controlled than a laboratory, introducing additional sources of distraction and inattention — they should suffice, and likely more so, in standard laboratory implementations. The practical implication extends naturally to certainty-equivalent choice lists used in risk attitude measurement, where the elicitation mechanism is comparable. One boundary condition worth exploring in future research is whether training intensity matters more when ambiguous outcomes derive from natural uncertainty sources, such as stock index variation, where respondents may have less intuitive footing than with the stylized urns studied here.

5 Conclusion

This paper studies whether increasing pre-task binary-choice training from 5 to 100 questions affects ambiguity-attitude elicitation via the matching-probability method. I find no evidence that more extensive training changes measured ambiguity attitudes. Extensive training does reduce stochastic-dominance violations when the risky option dominates the ambiguous one, but it leaves other indicators of response quality unchanged. In this setting, the standard five-question training module appears sufficient for ambiguity-attitude elicitation, while heavier training delivers only a limited compliance gain at the cost of greater respondent burden.

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